project Design Phase-I
Solution Architecture

Date	09 March 2026
Team ID	PNT2022TMIDxxxxxx
Project Name	Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau
Maximum Marks	4 Marks

Solution Architecture:

- Solution architecture is a structured framework that defines how different components of a system interact to solve a specific business problem.
- In this project, the solution architecture is designed to analyze housing market data and visualize trends using **Tableau dashboards integrated with a Flask web application**.
- The architecture bridges the gap between **raw housing data and meaningful insights** by combining data processing, visualization, and web deployment.
- The main goals of this architecture are:
- To collect and process housing market datasets.
- To perform data cleaning and preparation.
- To generate interactive visualizations using Tableau.
- To integrate dashboards with a web application.
- To provide users with insights about housing prices and property features.
- This architecture enables **real-time interaction with housing data through dashboards and storyboards**.
- Solution Architecture
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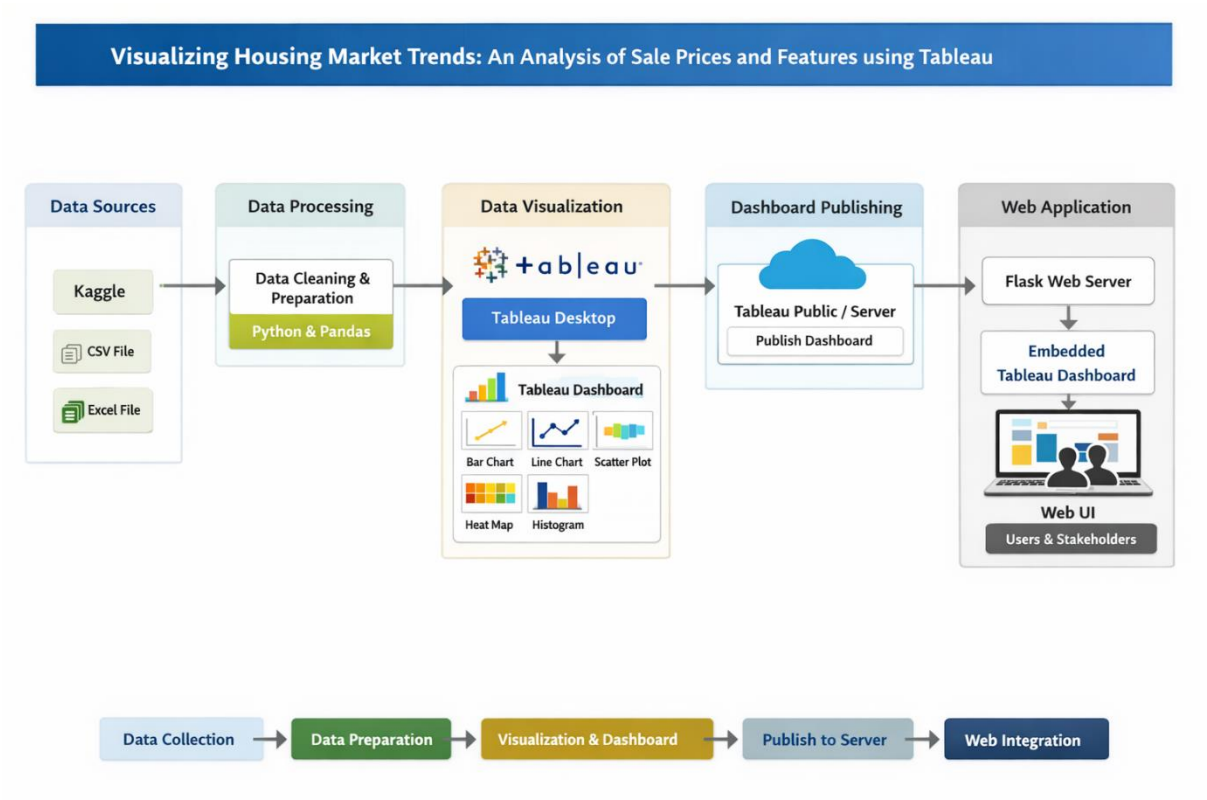
- **2. Components of the Solution Architecture**
- The proposed architecture consists of **five major layers**:
- Data Source Layer
- Data Processing Layer
- Visualization Layer
- Application Layer
- User Interface Laye
- and delivered.

Example - Solution Architecture Diagram:



4. Data Flow Architecture

The data flows through multiple stages from raw dataset to final visualization.

Step 1: Data Collection

Housing datasets are collected from sources such as:

- Kaggle
- UCI Machine Learning Repository
- Government housing databases

The dataset typically includes:

- Sale Price
- Lot Area
- Bedrooms
- Bathrooms
- Year Built
- Neighborhood

- Garage Size

The data is stored in CSV or Excel format.

Step 2: Data Preparation

Before visualization, the dataset undergoes data preprocessing.

Data preparation tasks include:

- Removing missing values
- Handling outliers
- Converting data types
- Renaming columns
- Feature engineering

Example calculated fields:

Price per Square Foot

Sale Price / Lot Area

House Age

Current Year - Year Built

Price Category

Low / Medium / High

This step ensures the dataset is ready for analysis.

Step 3: Data Modeling in Tableau

The cleaned dataset is connected to Tableau Desktop.

Tableau automatically categorizes data into:

Dimensions
Example:

- Neighborhood
- House Type

Measures
Example:

- Sale Price
- Lot Area

Calculated fields are created to derive additional insights.

Step 4: Data Visualization

Multiple interactive visualizations are created in Tableau.

Examples include:

Bar Chart
Average Sale Price by Neighborhood

Line Chart
Sale Price Trend by Year

Scatter Plot
Lot Area vs Sale Price

Heat Map
Neighborhood vs Average Price

Histogram
Distribution of Housing Prices

These visualizations help identify patterns and trends in housing markets.

Step 5: Dashboard Development

The visualizations are combined into a single interactive dashboard.

Dashboard features include:

- Interactive filters
- Tooltips
- Drill-down analysis
- Responsive layout

Users can filter data based on:

- Location
- Price range
- Number of bedrooms
- Year built

Step 6: Storyboard Creation

Tableau Story is used to present insights sequentially.

Example story flow:

1. Overview of housing market
2. Price comparison by neighborhood
3. Effect of house size on price
4. Price trend over time
5. Key insights and conclusions

This helps stakeholders understand the results easily.

Step 7: Dashboard Publishing

The dashboard is published using:

- Tableau Public
- Tableau Server
- Tableau Online

After publishing, Tableau generates an embed link.

Step 8: Web Integration using Flask

A Flask web application is created to embed the Tableau dashboard.

Flask handles:

- Web routing
- HTML page rendering
- Dashboard embedding

Example architecture:

Flask Server → HTML Page → Embedded Tableau Dashboard

This allows users to access the dashboard through a web interface.

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5. Technology Stack

Layer	Technology Used
Data Storage	CSV / Excel
Data Processing	Python, Pandas
Visualization	Tableau Desktop

Layer	Technology Used
Dashboard Hosting	Tableau Public
Web Integration	Flask
Frontend	HTML, CSS

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6. Performance Optimization

To ensure smooth performance, the following methods are used:

- Data extracts instead of live queries
- Optimized calculated fields
- Limited filters and visualizations
- Efficient data aggregation

These techniques improve dashboard responsiveness.

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7. Benefits of the Solution Architecture

The proposed architecture provides several advantages:

- Scalable data analysis
- Interactive visual exploration
- Easy deployment through web interface
- Clear understanding of housing market trends
- Support for decision-making in real estate markets

The architecture allows stakeholders to analyze housing price patterns and property features effectively.